Proceedings of the 16th International Conference on Finite or Infinite Dimensional Complex Analysis and Applications

http://16th.icfidcaa.org

July 28–August 1, 2008

Dongguk University

Gyeongju, Korea

Co-hosted by

- Dongguk University
- The Korean Mathematical Society
  (G-KMS Series-AN-08-0003)
- The Korean Federation of Science and Technology Societies

Supported by

- Gyeongju City
- Korea Hydro & Nuclear Power Co. LTD
  - Wol-Song Nuclear Power Site
  - Radwaste Disposal Facility Construction Site
- BK21 Dynamic Math Center

Edited by Junesang Choi, SoYoung Choi, Dae Ho Jin,
Seong-A Kim, and Jong Moon Shin

Published by Daeyang Printing, Gyeongju 780-180, Korea

ISBN 978-89-88678-15-2

May 5, 2009

Contents

History of the ICFIDCAA iv

Publications v

Topics of Conference vi

Committee vi

Addresses viii

List of Participants xii

Acknowledgements xx

Contributions 1

1. H. M. Srivastava, *Linearization relations for the Laguerre polynomials and the associated families of Dirichlet integrals* 1

2. Sui Sun Cheng, *Dual Sets of Envelopes and Characteristic Regions of Quasi-Polynomials* ... 12

3. Pei-Chu Hu and Chung-Chun Yang, *abc-conjectures and its analogues over function fields* .. 22

4. Tao Qian, *A new property of Nevanlinna functions* 38

5. Z. G. Feng, C. M. Liu, and K. L. Teo, *Optimal parameter selection of stochastic impulsive systems* 50

6. Hirotaka Akiyoshi, *Side parameter for the punctured torus groups* 58

7. Wolfram Bauer, *Toeplitz operators and enveloping algebras on spaces of entire and Pluriharmonic functions* 63

8. Yoshihiro Aihara, *Deficiencies of holomorphic curves in algebraic manifolds* 70

9. Z. X. Chen and K. H. Shon, *Zeros and fixed points of differences and divided differences for transcendental entire functions* 76

10. Dmitriy Khots and Boris Khots, *Tenth Hilbert problem in observer's mathematics* 81

11. *Ege Fujikawa and Katsuhiko Matsuzaki, The projection of limit sets of modular groups on asymptotic Teichmüller spaces* 86

12. K. Fujita, *Integral representations for holomorphic functions on some balls* 93

13. Hirofumi Goto and Hiroki Sato, *Remarks on Jørgensen groups of parabolic type* 99

14. Tatsuhiro Honda, *Estimates for some mappings on complex Banach spaces* 106

15. Shuechin Huang, *The signatures of torsion free subgroups in finitely generated Fuchsian groups* 114

16. Nak Eun Cho, *Sandwich-type theorems for a class nonlinear integral operators* 118

17. D. Karp, *Some bounds for the ratios of hypergeometric functions with shifted parameters* . 123

18. Victoria Kim, *Symmetrization-type transformation for Riemann surfaces* 132

19. D. Eelbode and L. Van de Voorde, *Higher spin operators and polyharmonic functions* 137

20. Yoichi Imayoshi and Toshihiro Nogi, *The number of holomorphic sections of a holomorphic family of Riemann surfaces induced by a certain Kodaira surface* 143

21. Masashi Kisaka, *On disconnected Julia sets of semihyperbolic transcendental entire functions* 148

22. E. G. Kwon, *On a secure weight* (1) *related to a weighted Bergman space* 154

23. Oh Sang Kwon, *Subordination properties of multivalent functions* 159

24. Makoto Masumoto, *Conformal mappings of once-holed tori into Riemann surfaces of positive genus* 166

25. Masaru Nishihara, *Invariant holomorphic functions for the complex general linear group action on Banach spaces* 172

26. Katsuhiko Matsuzaki and Yasuhiro Yabuki, *An example of self-covering of Riemann surface* 184

27. Yoshihiro Mizuta, Sobolev's *Inequality for Riesz potentials with variable exponents approaching* 189

28. S. Nakane, *Remarks on Axiom A polynomial skew products of higher degrees* 196

29. Ekrem Savaş, *Generalization of asymptotically probability equivalent sequences* 201

30. T. N. Shanmugam, *Fekete-Szegö problems in the geometric function theory* 206

31. Yilmaz Simsek, *p-adic Dedekind and Hardy-Berndt sums related to Volkenborn integral on* $\mathbb{Z}_p$ 222

32. M. Kh. Ruziev, *The Dirichlet problem in the vertical half-band for the degenerated elliptic equation with singular coefficient* 232

33. Shoji Tsuboi, *Linear projections of rational threefolds* 237

34. Mamasali Turaev, *Decomposition formulas for the product of two Clausen functions involving Kampe de Feriet functions* 248

35. A. Hasanov, *Applications of an operator $H(\alpha, \beta)$ to the Lauricella multivariable hypergeometric functions* 267

36. A. K. Urinov and I. T. Tojiboev, *Eigenvalue and eigenfunctions of some boundary-value problems for a mixed type equation with non-smooth line of type changing* 282

37. Chi Song Wong, *Moments and characterizations of complex normal and Wishart distributions* 290

38. Zhaojun Wu, *Angular distribution and complex oscillation* 297

39. M. Nishio, N. Suzuki, and M. Yamada, *Toeplitz operators of Schatten class on the parabolic Bergman space* 305

Some Bounds for the Ratios of Hypergeometric Functions with Shifted Parameters

D. Karp

Institute of Applied Mathematics
7 Radio St., Vladivostok, 690041, Russia
E-Mail: **dmkrp@yandex.ru**

Abstract. We demonstrate two approaches to finding bounds for the ratios of hypergeometric functions and studying their monotonicity. One is based on generalized Stieltjes transform representation, the other uses some recent log-concavity results for series. Besides, a conjecture asserting the monotonicity of a quotient of products of confluent hypergeometric functions is made.

1. Introduction

We will use the standard notation for the generalized hypergeometric function:

$$ {}_pF_q\left((a_p);(b_q);z\right) = {}_pF_q\left(\begin{matrix}(a_p)\\(b_q)\end{matrix}\,\middle|\, z\right) = \sum_{n=0}^{\infty} \frac{(a_1)_n (a_2)_n \cdots (a_p)_n}{(b_1)_n \cdots (b_q)_n n!} z^n, \tag{1} $$

where $(a_p) = a_1, a_2, \ldots, a_p$ is the parameter list and $(a)_n = a(a+1)\cdots(a+n-1)$ is the shifted factorial. The series converges for all finite z if $p \le q$ and for $|z| < 1$ when $p = q+1$. In the latter case it can be analytically continued onto the entire complex plane cut along $[1, \infty)$. The celebrated continued fraction of Gauss

$$ \frac{{}_2F_1(a+1, b; c+1; -x)}{{}_2F_1(a, b; c; -x)} = \frac{c}{c} + \frac{b(c-a)x}{c+1} + \frac{(a+1)(c+1-b)x}{c+2} + \cdots + $$
$$ + \frac{(b+n)(c-a+n)x}{c+2n+1} + \frac{(a+1+n)(c+1-b+n)x}{c+2n+2} + \cdots \tag{2} $$

converges and has positive elements when $x > 0$ and $c > a > -1$, $c+1 > b > 0$, $c > 0$. Hence, its even convergents form an increasing sequence and approximate the left-hand side from below, while the odd convergents form a decreasing sequence and approximate the left-hand side from above (see details in [4, 6, 10]). Taking the first two terms we get

$$ \frac{c(c+1)}{c(c+1) + b(c-a)x} < \frac{{}_2F_1(a+1, b; c+1; -x)}{{}_2F_1(a, b; c; -x)} < 1. \tag{3} $$

---

[1]2000 *Mathematics Subject Classification.* Primary 33C20, 33C15, 33C05.

[2]*Key words and phrases.* Hypergeometric function, Kummer function, Gauss function

[3]This work has been supported by the Far Eastern Branch of the Russian Academy of Sciences (grant 09-III-A-01-008), Russian Basic Research Fund (grant 08-01-00028-a) and Presidential Grant for Leading Scientific Schools (grant 2810.2008.1).

Both bounds are clearly sharp at $x = 0$. To see what happens around $x = \infty$ we can use the formula [1, Theorem 2.3.2]

$$
\begin{aligned}
{}_2F_1(a,b;c;-x) &= C(a,b,c)x^{-a}{}_2F_1(a,a-c+1;a-b+1;-1/x) \\
&\qquad + D(a,b,c)x^{-b}{}_2F_1(b,b-c+1;b-a+1;-1/x),
\end{aligned}
$$

$$
C(a,b,c) = \frac{\Gamma(c)\Gamma(b-a)}{\Gamma(c-a)\Gamma(b)},\ D(a,b,c) = \frac{\Gamma(c)\Gamma(a-b)}{\Gamma(c-b)\Gamma(a)},
$$

which expresses the fact that any three solutions of hypergeometric differential equation must be linearly related, but can also be viewed as the asymptotic expansion around $x = \infty$. Using this formula we see that

$$
\frac{{}_2F_1(a+1,b;c+1;-x)}{{}_2F_1(a,b;c;-x)} = \begin{cases} \frac{C(a+1,b,c+1)}{C(a,b,c)x}(1+o(1)),\ b > a+1, \\ \frac{D(a+1,b,c+1)}{C(a,b,c)x^{b-a}}(1+o(1)),\ a < b < a+1, \\ \frac{D(a+1,b,c+1)}{D(a,b,c)}(1+o(1)),\ b < a. \end{cases}
$$

Hence, the lower bound in (3) gives the correct value of the ratio at infinity if $b > a$ and the correct rate of decrease if $b > a + 1$ but never the correct constant. Many other continued fractions for the ratios of the Gauss functions, the confluent hypergeometric functions and the ${}_3F_2$ hypergeometric functions are found in [4].

For $x \in (0,1)$ the monotonicity properties of ${}_2F_1(A,B;C;x)/{}_2F_1(a,b;c;x)$ for positive A, B, C, a, b and c have been thoroughly studied in [13, Theorem 2.31]. We also mention the paper by Bordelon [2] who investigated the bounds for ${}_1F_1(a,c,x)/{}_1F_1(a,d,x)$ and ${}_1F_1(a,c,x)/{}_1F_1(a,c,y)$ and the work of Joshi and Nalwaya [7] where two-sided bounds for the modified Struve function (expressible in terms of ${}_1F_2$) and their ratios are established.

Further bounds for ratios of hypergeometric functions can be obtained by using the two-sided inequalities for the Gauss hypergeometric function given in [3] or the generalized hypergeometric function given in [11]. This method, however, does not produce very precise estimates, since one has to divide the upper bound by the lower bound and vice versa which leads inevitably to some loss of precision.

In this note we will demonstrate two more approaches to studying monotonicity of and obtaining the bounds for the ratios of the generalized hypergeometric functions (1). One is based on generalized Stieltjes transform representation satisfied by ${}_{q+1}F_q$ when its denominator parameters majorize those of numerator introduced in [8] and the other is related to log-concavity in parameters of functions defined as power series whose coefficients involve shifted factorials of these parameters. The latter approach has been recently investigated in [9].

2. Generalized Stieltjes representation and its consequences.

We begin with a representation of ${}_{q+1}F_q(z)$ as the generalized Stieltjes transform.

Theorem 1. *For $\Re b_k > \Re a_k > 0$, $k = 1,2,\dots,q$, and $|\arg(1+z)| < \pi$ we have*

$$
{}_{q+1}F_q\left(\begin{matrix}\sigma,(a_q)\\(b_q)\end{matrix}\,\middle|\,-z\right) = A_0\int_0^1 \frac{s^{a_1-1}g((a_q);(b_q);s)ds}{(1+sz)^\sigma}, \tag{4}
$$

where

$$g((a_q);(b_q);s)=\int\limits_{\Lambda_q(s)}[1-s/(t_2\cdots t_q)]^{b_1-a_1-1}\prod_{k=2}^{q}t_k^{a_k-a_1-1}(1-t_k)^{b_k-a_k-1}\,dt_2\cdots dt_q, \quad (5)$$

$$\Lambda_q(s)=[0,1]^{q-1}\cap\{t_2,\ldots,t_q:\ t_2\cdots t_q>s\} \quad (6)$$

and

$$A_0=\Gamma\left[\begin{matrix}(b_q)\\(a_q),(b_q-a_q)\end{matrix}\right]\equiv\frac{\Gamma(b_1)\cdots\Gamma(b_q)}{\Gamma(a_1)\cdots\Gamma(a_q)\Gamma(b_1-a_1)\cdots\Gamma(b_q-a_q)}. \quad (7)$$

The function $g((a_q);(b_q);s)$ is invariant under simultaneous shift of all parameters:

$$g((a_q)+\delta;(b_q)+\delta;s)=g((a_q);(b_q);s) \quad (8)$$

for any complex δ.

Proof. The multiple Euler integral representation for ${}_{q+1}F_q(z)$ reads [14, formula 7.2.3(10)]:

$${}_{q+1}F_q\left(\begin{matrix}\sigma,(a_q)\\(b_q)\end{matrix}\,\middle|\,-z\right)=\Gamma\left[\begin{matrix}(b_q)\\(a_q),(b_q-a_q)\end{matrix}\right]\int\limits_{[0,1]^q}\frac{\prod_{k=1}^{q}t_k^{a_k-1}(1-t_k)^{b_k-a_k-1}}{(1+t_1t_2\ldots t_qz)^{\sigma}}\,dt_1\cdots dt_q. \quad (9)$$

Integration in (9) is over the q-dimensional unit cube, $\Re b_k>\Re a_k>0$ for all $k=1,2,\ldots,q$, and $|\arg(1+z)|<\pi$. Formula (9) is obtained by repeated application of the generalized Euler integral [1, formula (2.2.2)]

$${}_{q+1}F_q\left(\begin{matrix}(a_q),a_{q+1}\\(b_{q-1}),b_q\end{matrix}\,\middle|\,z\right)=\frac{\Gamma(b_q)}{\Gamma(a_{q+1})\Gamma(b_q-a_{q+1})}\int\limits_0^1 t^{a_{q+1}-1}(1-t)^{b_q-a_{q+1}-1}\,{}_qF_{q-1}\left(\begin{matrix}(a_q)\\(b_{q-1})\end{matrix}\,\middle|\,zt\right)dt,$$

where the last step with $q=1$ is the standard Euler integral for ${}_2F_1$.

We make the variable change $s=t_1t_2\cdots t_q$, $t_1=s/(t_2\cdots t_q)$, leaving $t_2,\ldots,t_q$ unchanged. The Jacobian of such transformation is

$$J(s,t_2,\ldots,t_q)=\frac{1}{t_2\cdots t_q}.$$

Hence, we get from (9):

$$\begin{aligned}{}_{q+1}F_q\left(\begin{matrix}\sigma,(a_q)\\(b_q)\end{matrix}\,\middle|\,-z\right)&=A_0\int\limits_0^1\frac{ds}{(1+sz)^{\sigma}}\\&\times\int\limits_{\Lambda_q(s)}[s/(t_2\cdots t_q)]^{a_1-1}[1-s/(t_2\cdots t_q)]^{b_1-a_1-1}\prod_{k=2}^{q}t_k^{a_k-1}(1-t_k)^{b_k-a_k-1}\,\frac{dt_2\cdots dt_q}{t_2\cdots t_q}\\&=A_0\int\limits_0^1\frac{s^{a_1-1}ds}{(1+sz)^{\sigma}}\int\limits_{\Lambda_q(s)}[1-s/(t_2\cdots t_q)]^{b_1-a_1-1}\prod_{k=2}^{q}t_k^{a_k-a_1-1}(1-t_k)^{b_k-a_k-1}\,dt_2\cdots dt_q,\end{aligned}$$

where $\Lambda_q(s)$ and A_0 are defined in (6) and (7), respectively. Introducing the notation (5) for the inner integral we arrive at formula (4). The shift invariance (8) is obvious from the definition of $g((a_q);(b_q);s)$. $\square$

Combining representation (4) with the Chebyshev inequality we get two corollaries dealing with two different types of monotonic behavior.

Corollary 1.1. *Put*

$$g(\sigma,x)={}_{q+1}F_q\left(\begin{matrix}\sigma,(a_q)\\(b_q)\end{matrix}\,\middle|\,-x\right)$$

Suppose $b_i>a_i>0$, $i=1,\dots,q$, and $x>-1$. Then the function $\sigma\mapsto g(\sigma+\delta,x)/g(\sigma,x)$ is monotone increasing on $[0,\infty)$ for any fixed $\delta>0$.

Remark 1. Functions $g(y)$ with monotone increasing $g(y+\delta)/g(y)$ for any positive δ are called Wright log-convex. For continuous functions Wright log-convexity is equivalent to usual log-convexity (which is not true in general). See details in [12, p.3].

Proof. The claimed monotonicity is the inequality

$$g(\sigma+\delta,x)g(\beta,x)<g(\beta+\delta,x)g(\sigma,x)$$

for $\beta>\sigma>0$. Writing representation (4) for $g(\sigma,x)$ we see that this inequality is precisely the Chebyshev inequality [12, Chapter IX, formula (1.1)]

$$\int_0^1 q(s)p(s)ds\int_0^1 h(s)p(s)ds<\int_0^1 q(s)h(s)p(s)ds\int_0^1 p(s)ds, \tag{10}$$

with

$$p(s)=\frac{s^{a_1-1}g((a_q);(b_q);s)}{(1+sx)^{\sigma}},\ q(s)=\frac{1}{(1+sx)^{\delta}},\ h(s)=\frac{1}{(1+sx)^{\beta-\sigma}}.$$

The functions $q(s)$ and $h(s)$ are either both increasing or both decreasing (depending on the sign of x) and $p(s)>0$ which implies the validity of the Chebyshev inequality. $\square$

Corollary 1.2. *Suppose $\delta>0$, $b_k>a_k>0$, $k=1,\dots,q$. The function*

$$f(\sigma,(a_q);(b_q);\delta;x)\equiv\frac{{}_{q+1}F_q\left(\sigma,(a_q)+\delta;(b_q)+\delta;-x\right)}{{}_{q+1}F_q\left(\sigma,(a_q);(b_q);-x\right)} \tag{11}$$

is monotone decreasing if $\sigma>0$ and monotone increasing if $\sigma<0$ for all $x>-1$.

Proof. Put

$$A_\delta=\Gamma\left[\begin{matrix}(b_q)+\delta\\(a_q)+\delta,(b_q-a_q)\end{matrix}\right].$$

Then by (4) and (8):

$$f(\sigma,(a_q);(b_q);\delta;x)=\frac{A_\delta\int_0^1 s^{a_1+\delta-1}(1+sx)^{-\sigma}g((a_q)+\delta;(b_q)+\delta;s)ds}{A_0\int_0^1 s^{a_1-1}(1+sx)^{-\sigma}g((a_q);(b_q);s)ds}$$

$$=\frac{A_\delta\int_0^1 s^{a_1+\delta-1}(1+sx)^{-\sigma}g((a_q);(b_q);s)ds}{A_0\int_0^1 s^{a_1-1}(1+sx)^{-\sigma}g((a_q);(b_q);s)ds}.$$

The statement of the theorem is equivalent to saying that $f'_x(\sigma,(a_q);(b_q);\delta;x)<0$ when $\sigma>0$ and $f'_x(\sigma,(a_q);(b_q);\delta;x)>0$ when $\sigma<0$. Differentiating the definition of f above we see that both inequalities are equivalent to the single inequality (10) with

$$p(s)=s^{a_1+\delta-1}(1+sx)^{-\sigma}g((a_q);(b_q);s),$$

$$q(s)=s^{\delta},\ h(s)=\frac{s}{1+xs}.$$

The function $p(s)$ is positive, while the functions $q(s)$ and $h(s)$ are monotone increasing for fixed $x>-1$ and $0<s<1$. Hence, the above inequality is an instance of the Chebyshev inequality (10). □

The value of $f(\sigma,(a_q);(b_q);\delta;x)$ at $x=\infty$ can be found using the representation of ${}_{q+1}F_q(z)$ in the neighborhood of the singular point $z=\infty$. Assume for the moment that no numerator parameters differ by an integer. Then, according to [14, formula 7.2.3.77],

$$\begin{aligned}{}_{q+1}F_q\left(\begin{matrix}(a_{q+1})\\(b_q)\end{matrix}\,\middle|\,-z\right)=&\Gamma\left[\begin{matrix}(b_q)\\(a_{q+1})\end{matrix}\right]\sum_{k=1}^{q+1}\Gamma\left[\begin{matrix}a_k,(a_{q+1})'-a_k\\(b_q)-a_k\end{matrix}\right]\\&\cdot{}_{q+1}F_q\left(\begin{matrix}1+a_k-(b_q),a_k\\1+a_k-(a_{q+1})'\end{matrix}\,\middle|\,-1/z\right)z^{-a_k},\end{aligned}\tag{12}$$

where the prime at (a_{q+1}) means that the term a_k is excluded from the list. It follows that for $\sigma<\min(a_1,a_2,\dots,a_q)$ we will have

$${}_{q+1}F_q\left(\begin{matrix}\sigma,(a_q)\\(b_q)\end{matrix}\,\middle|\,-z\right)=\Gamma\left[\begin{matrix}(a_q)-\sigma,(b_q)\\(a_q),(b_q)-\sigma\end{matrix}\right]z^{-\sigma}+o(z^{-\sigma}),\ z\to\infty.\tag{13}$$

Hence,

$$f(\sigma,(a_q);(b_q);\delta;\infty)=\Gamma\left[\begin{matrix}(a_q),(a_q)+\delta-\sigma,(b_q)-\sigma,(b_q)+\delta\\(a_q)-\sigma,(a_q)+\delta,(b_q),(b_q)+\delta-\sigma\end{matrix}\right].\tag{14}$$

When some of the numerator parameters on the left-hand side of (12) differ by an integer, formula (12) breaks down and one has to resort to much more involved [14, formula 7.2.3.78]. Formula (14), however, remains valid by continuity (${}_pF_q/\Gamma[(b_q)]$ is an entire function of its parameters - see [14, 7.3.2.8]).

Formula (14) and Corollary 1.2 imply the following inequalities valid for $x>0$:

$$f(\sigma,(a_q);(b_q);\delta;\infty)<f(\sigma,(a_q);(b_q);\delta;x)<1=f(\sigma,(a_q);(b_q);\delta;0),\tag{15}$$

for positive σ and

$$1<f(\sigma,(a_q);(b_q);\delta;x)<f(\sigma,(a_q);(b_q);\delta;\infty),\tag{16}$$

for negative σ.

3. Series representation and its consequences

The following quite general result for series proved in [9] gives rise to two-sided inequalities for hypergeometric functions.

Theorem 2. *Let $f(a,x)$ be given by*

$$f(a,x)=\sum_{n=0}^{\infty} f_n(a)_n\frac{x^n}{n!}, \tag{17}$$

where $f_n>0$ (and is independent of a) and the series converges in some disk $|x|<R$. Suppose $\{f_n/f_{n-1}\}$ is decreasing. Then for $b>a>0$, $\delta>0$ and $0<x<R$:

$$\frac{\Gamma(a+\delta)\Gamma(b)}{\Gamma(b+\delta)\Gamma(a)}<\frac{f(b+\delta,x)f(a,x)}{f(a+\delta,x)f(b,x)}<1. \tag{18}$$

Remark 2. Choosing $b=a+\delta$ and $\delta=1$ we get the direct and reverse Turán type inequalities for $f(a,x)$:

$$\frac{a}{a+1}<\frac{f(a+2,x)f(a,x)}{f(a+1,x)^2}<1.$$

When applying Theorem 2 to hypergeometric functions we can combine it with hypergeometric transformations to extend the validity of (18) to negative x. For example, for the Kummer function ${}_1F_1(a;c;x)$ we get

Corollary 2.1. *For $b>a>0$ and $c,x>0$:*

$$\frac{\Gamma(a+\delta)\Gamma(b)}{\Gamma(b+\delta)\Gamma(a)}<\frac{{}_1F_1(b+\delta;c;x){}_1F_1(a;c;x)}{{}_1F_1(a+\delta;c;x){}_1F_1(b;c;x)}<1; \tag{19}$$

for $a<b<c-\delta$ and $c>0>x$:

$$\frac{\Gamma(c-a-\delta)\Gamma(c-b)}{\Gamma(c-b-\delta)\Gamma(c-a)}<\frac{{}_1F_1(b+\delta;c;x){}_1F_1(a;c;x)}{{}_1F_1(a+\delta;c;x){}_1F_1(b;c;x)}<1, \tag{20}$$

both sides of both inequalities are sharp in the sense that the upper bound is attained at $x=0$ and the lower bounds in (19), (20) *are attained at $x=+\infty$ and $x=-\infty$, respectively.*

Proof. Since $f_n/f_{n-1}=1/(c+n-1)$ is decreasing we are in the position to apply Theorem 2 which ensures the validity of (19). To prove (20) apply the Kummer transformation ${}_1F_1(a;c;x)=e^{-x}{}_1F_1(c-a;c;-x)$. The sharpness of (19) and (20) at infinity is seen from [1, Corollary 4.2.3]. □

Numerical tests suggests that the following enhancement of Corollary 2.1 is true:

Conjecture. The ratio in the middle of (19) is monotone decreasing from $(0,\infty)$ onto $(1,A)$, where A is the left-hand side of (19); the ratio in the middle of (20) is monotone increasing from $(-\infty,0)$ onto $(B,1)$, where B is the left-hand side of (20).

For the Gauss function ${}_2F_1(a,b;c;x)$ we have

Corollary 2.2. *For $0<x<1$:*

$$\frac{\Gamma(a+\delta)\Gamma(a')}{\Gamma(a)\Gamma(a'+\delta)}<\frac{{}_2F_1(a'+\delta,b;c;x){}_2F_1(a,b;c;x)}{{}_2F_1(a+\delta,b;c;x){}_2F_1(a',b;c;x)}<1 \tag{21}$$

if $b>c>0$, $a'>a>0$ and

$$\frac{\Gamma(c-a-\delta)\Gamma(c-a')}{\Gamma(c-a)\Gamma(c-a'-\delta)}<\frac{{}_2F_1(a'+\delta,b;c;x){}_2F_1(a,b;c;x)}{{}_2F_1(a+\delta,b;c;x){}_2F_1(a',b;c;x)}<1 \tag{22}$$

if $c > 0 > b$, $c - \delta > a' > a$. *Further for* $x < 0$ (21) *holds true if* $c > 0 > b$, $a' > a > 0$ *and* (22) *holds true if* $b > c > 0$, $c - \delta > a' > a$.

Proof. If $b > c > 0$, then the sequence

$$\frac{(b)_{k+1}/(c)_{k+1}}{(b)_k/(c)_k} = \frac{b+k}{c+k},\ k = 0, 1, 2, \ldots,$$

is decreasing and we are in the position to apply Theorem 2 which ensures the validity of (21) for $0 < x < 1$. Inequality (22) and all statements pertaining to negative x follow from (21) by an application of Euler's and Pfaff's transformations:

$$ {}_2F_1(a, b; c; x) = (1-x)^{c-a-b}\,{}_2F_1(c-a, c-b; c; x)$$

$$= (1-x)^{-a}\,{}_2F_1(a, c-b; c; x/(x-1)) = (1-x)^{-b}\,{}_2F_1(c-a, b; c; x/(x-1)).\ \square$$

Application of Theorem 2 to generalized hypergeometric function ${}_{q+1}F_q$ hinges on the following observation which might be of independent interest:

Lemma 1. *Let* $A(x) = a_0 + a_1 x + \cdots + a_n x^n$ *and* $B(x) = b_0 + b_1 x + \cdots + b_n x^n$ *have positive coefficients. Then* $A'(x)B(x) - B'(x)A(x)$ *has non-negative coefficients if*

$$\frac{a_n}{b_n} \geq \frac{a_{n-1}}{b_{n-1}} \geq \cdots \geq \frac{a_1}{b_1} \geq \frac{a_0}{b_0} \tag{23}$$

and non-positive coefficients if

$$\frac{a_n}{b_n} \leq \frac{a_{n-1}}{b_{n-1}} \leq \cdots \leq \frac{a_1}{b_1} \leq \frac{a_0}{b_0}. \tag{24}$$

If A *and* B *are not identical then some of the coefficients are positive under* (23) (*so that* $A'(x)B(x) - B'(x)A(x) > 0$ *for* $x > 0$) *and negative under* (24) (*so that* $A'(x)B(x) - B'(x)A(x) < 0$ *for* $x > 0$).

Proof. We have

$$A'(x)B(x) - B'(x)A(x) = \sum_{k=1}^{n}\sum_{i=0}^{n} k a_k b_i x^{i+k-1} - \sum_{k=1}^{n}\sum_{i=0}^{n} k b_k a_i x^{i+k-1}$$

$$= \frac{1}{x}\sum_{k=1}^{n}\sum_{i=0}^{n} x^{i+k} k(a_k b_i - a_i b_k) = \sum_{m=1}^{2n} x^{m-1} \sum_{\substack{i+k=m \\ i\geq 0, k\geq 1}}^{k,i\leq n} k(a_k b_i - a_i b_k)$$

Since each term in the inner sum with $i = k$ is clearly zero, we may write:

$$\sum_{\substack{i+k=m \\ i\geq 0, k\geq 1}}^{k,i\leq n} k(a_k b_i - a_i b_k) = \sum_{\substack{i+k=m, k<i \\ i\geq 0, k\geq 1}}^{k,i\leq n} k(a_k b_i - a_i b_k) + \sum_{\substack{i+k=m, k>i \\ i\geq 0, k\geq 1}}^{k,i\leq n} k(a_k b_i - a_i b_k)$$

Due to condition (23) every term in the second sum is non-negative. For each term in the first sum (say indexed $k = k^*$, $i = i^*$, $k^* < i^*$), there is a term in the second sum with $k = i^*$, $i = k^*$ and

$$k^*(a_{k^*} b_{i^*} - a_{i^*} b_{k^*}) + i^*(a_{i^*} b_{k^*} - a_{k^*} b_{i^*}) = (i^* - k^*)(a_{i^*} b_{k^*} - a_{k^*} b_{i^*}) \geq 0$$

Since A and B are not identical at least one of the inequalities (23) is strict which implies that some of the terms above is strictly positive. The second statement follows

from $A'(x)B(x) - B'(x)A(x) = -(B'(x)A(x) - A'(x)B(x))$ by exchanging the roles of A and B. □

Remark 3. It is easy to verify directly that conditions (23) are also necessary for $A'(x)B(x) - B'(x)A(x) > 0$ if $x > 0$ when $n = 1$ and $n = 2$ (similarly for conditions (24)).

Remark 4. After this paper was already finished Árpád Baricz brought the recently published article [5] to our attention. Lemma 1 repeats almost precisely the contents of Theorem 4.4 from [5]. Interestingly, the proof in [5] is by induction and hence differs substantially from ours.

Now consider

$$R(x) = \frac{\prod_{k=1}^{q}(a_k + x)}{\prod_{k=1}^{q}(b_k + x)}$$

with positive a_k, b_k. Let $e_m(c_1, \ldots, c_q)$ denote m-th elementary symmetric polynomial,

$$e_1(c_1, \ldots, c_q) = c_1 + c_2 + \cdots + c_q,$$

$$e_2(c_1, \ldots, c_q) = c_1c_2 + c_1c_3 + \cdots + c_1c_q + c_2c_3 + c_2c_4 + \cdots + c_2c_q + \cdots + c_{q-1}c_q,$$

$$\ldots,$$

$$e_q(c_1, \ldots, c_q) = c_1c_2 \cdots c_q.$$

Lemma 2. *The function $R(x)$ is monotone increasing on $(0, \infty)$ if*

$$\frac{e_q(b_1, \ldots, b_q)}{e_q(a_1, \ldots, a_q)} \geq \frac{e_{q-1}(b_1, \ldots, b_q)}{e_{q-1}(a_1, \ldots, a_q)} \geq \cdots \geq \frac{e_1(b_1, \ldots, b_q)}{e_1(a_1, \ldots, a_q)} \geq 1 \tag{25}$$

and monotone decreasing if

$$\frac{e_q(b_1, \ldots, b_q)}{e_q(a_1, \ldots, a_q)} \leq \frac{e_{q-1}(b_1, \ldots, b_q)}{e_{q-1}(a_1, \ldots, a_q)} \leq \cdots \leq \frac{e_1(b_1, \ldots, b_q)}{e_1(a_1, \ldots, a_q)} \leq 1. \tag{26}$$

Proof. We have

$$A(x) \equiv \prod_{k=1}^{q}(a_k + x) = \sum_{k=0}^{q} e_k(a_1, \ldots, a_q)x^{q-k} = \sum_{k=0}^{q} e_{q-k}(a_1, \ldots, a_q)x^k$$

$$B(x) \equiv \prod_{k=1}^{q}(b_k + x) = \sum_{k=0}^{q} e_k(b_1, \ldots, b_q)x^{q-k} = \sum_{k=0}^{q} e_{q-k}(b_1, \ldots, b_q)x^k$$

$$R'(x) = \frac{A'(x)B(x) - B'(x)A(x)}{[B(x)]^2}$$

Hence by Lemma 1 we can assert that

$$1 = \frac{e_0(a_1, \ldots, a_q)}{e_0(b_1, \ldots, b_q)} \geq \frac{e_1(a_1, \ldots, a_q)}{e_1(b_1, \ldots, b_q)} \geq \frac{e_2(a_1, \ldots, a_q)}{e_2(b_1, \ldots, b_q)} \geq \cdots \geq \frac{e_q(a_1, \ldots, a_q)}{e_q(b_1, \ldots, b_q)}$$

which is the same as (25), is sufficient for $R'(x) > 0$. Similarly for $R'(x) < 0$. □

Corollary 2.3. *Suppose* $a_i, b_i > 0$, $i = 1, \ldots, q$, $\beta > \sigma > 0$ *and condition* (26) *is satisfied. Then for* $x \in (0,1)$

$$\frac{\Gamma(\sigma+\delta)\Gamma(\beta)}{\Gamma(\beta+\delta)\Gamma(\sigma)} < \frac{{}_{q+1}F_q\left(\beta+\delta,(a_q);(b_q);x\right){}_{q+1}F_q\left(\sigma,(a_q);(b_q);x\right)}{{}_{q+1}F_q\left(\sigma+\delta,(a_q);(b_q);x\right){}_{q+1}F_q\left(\beta,(a_q);(b_q);x\right)} < 1.$$

Remark 5. For ${}_3F_2\left(\sigma,a_1,a_2;b_1,b_2;x\right)$ conditions (26) read:

$$\frac{b_1 b_2}{a_1 a_2} \le \frac{b_1+b_2}{a_1+a_2} \le 1.$$

According to remark 3 after Lemma 1 in this case these conditions are both necessary and sufficient for decrease of the function $x \mapsto (a_1+x)(a_2+x)/[(b_1+x)(b_2+x)]$.

The claim of Lemma 1 made under condition (24) clearly remains true when $a_n = a_{n-1} = a_{n-r} = 0$. Condition (26) then reads

$$\frac{e_q(b_1,\ldots,b_q)}{e_{q-r-1}(a_1,\ldots,a_{q-r-1})} \le \frac{e_{q-1}(b_1,\ldots,b_q)}{e_{q-r-2}(a_1,\ldots,a_{q-r-1})} \le \cdots \le \frac{e_{r+2}(b_1,\ldots,b_q)}{e_1(a_1,\ldots,a_{q-r-1})} \le e_{r+1}(b_1,\ldots,b_q)$$

This permits the application of Theorem 2 to ${}_pF_q$ with $p \le q$. For instance, for ${}_2F_2(\sigma,a_1;b_1,b_2;x)$ Theorem 2 is true if $a_1 \ge b_1 b_2/(b_1+b_2)$.

References

[1] G. E. Andrews, R. Askey and R. Roy, *Special functions*, Cambridge University Press, 1999.

[2] D. J. Bordelon, Inequalities for special functions Problem 72-15 (proposed by D.K. Ross), *SIAM Review* **15**(3) (1973), 665-670.

[3] B. C. Carlson, Some inequalities for hypergeometric functions, *Proc. Amer. Math. Soc.* **17**(1) (1966), 32-39.

[4] A. Cuyt, V. B. Petersen, B. Verdonk, H. Waadeland, W.B. Jones, *Handbook of Continued Fractions for Special Functions*, Springer, 2008.

[5] V. Heikkala, M. K. Vamanamurthy, M. Vuorinen, Generalized Elliptic Integrals, *Computational Methods and Function Theory* **9**(1) (2009), 75-109.

[6] P. Henrici, *Applied and Computational Complex Analysis*, Vol.2, Wiley, 1977.

[7] C. M. Joshi and S. Nalwaya, Inequalities for modified Struve functions, *J. Indian Math. Soc.* **65** (1998) 49-57.

[8] D. Karp and S. M. Sitnik, Inequalities and monotonicity of ratios for generalized hypergeometric function, *J. Approx. Theo.*, 2008 (article in press), doi:10.1016/j.jat.2008.10.002.

[9] D. Karp and S. M. Sitnik, Log-convexity and log-concavity of hypergeometric-like functions, Preprint.

[10] L. Lorentzen and H. Waadeland, *Continued Fractions with Applications*, North-Holland, 1992.

[11] Y. L. Luke, Inequalities for generalized hypergeometric functions, *J. Approx. Theo.* **5** (1972), 41-65.

[12] D. S. Mitrinović, J. E. Pecarić, A. M. Fink, *Classical and New Inequalities in Analysis*, Kluwer Academic Publishers, 1993.

[13] S. Ponnusamy and M. Vuorinen, Asymptotic expansions and inequalities for hypergeometric functions, *Mathematika* **44** (1997), 278-301.

[14] A. P. Prudnikov, Yu. A. Brychkov, O. I. Marichev, *Integrals and Series*, Vol.3, Additional Chapters, Moscow Nauka, 1986. English translation: Integrals and series. Vol. 3: More special functions. New York: Gordon and Breach Science Publishers, 1990.